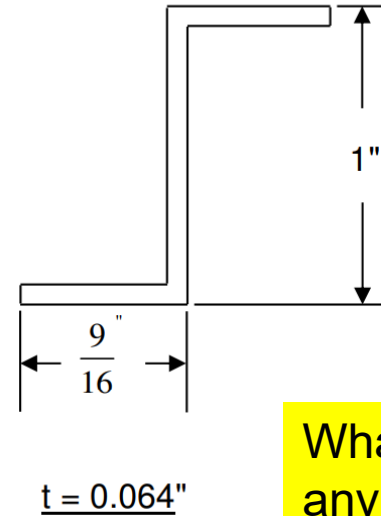
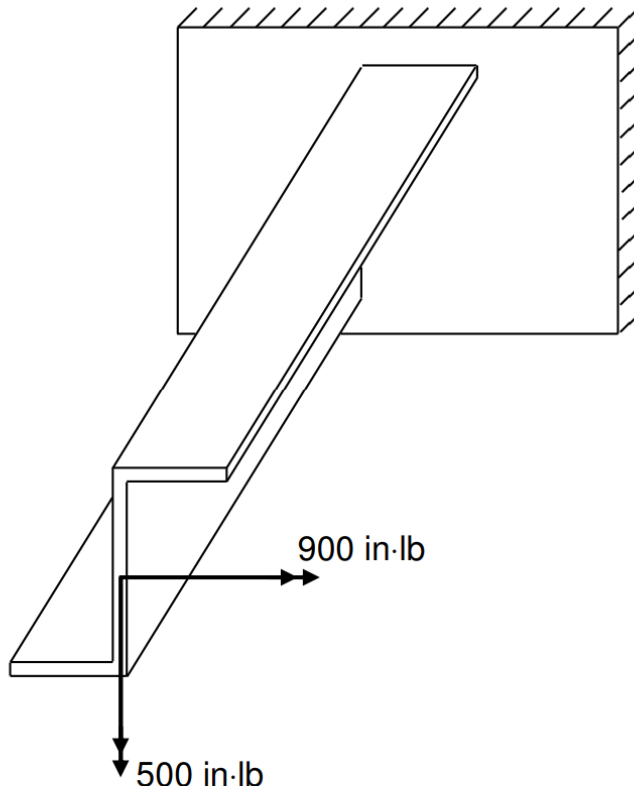


4.1: General Flexure Formula

Consider the cantilevered z-beam below with two bending couples as applied loads. Z-beams are often used as stiffeners in aircraft construction with the upper or lower surface riveted to the skin.



What symmetry (if any) does this cross section have?

Find

- (a) the normal stress variation over the cross-section and
- (b) the equation of the neutral axis.
- (c) Sketch the normal stress distribution.



4.1: General Flexure Formula

- Essentially, part (a) is asking us to determine M_y , M_z , I_y , I_z , and I_{yz} and insert those into the general equation:

$$\sigma_x = \frac{-y(I_y M_z + I_{yz} M_y) + z(I_{yz} M_z + I_z M_y)}{I_y I_z - I_{yz}^2}$$

so that we will have an expression for σ_x in terms of y and z

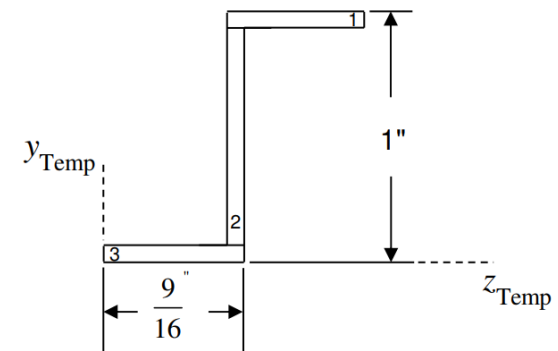
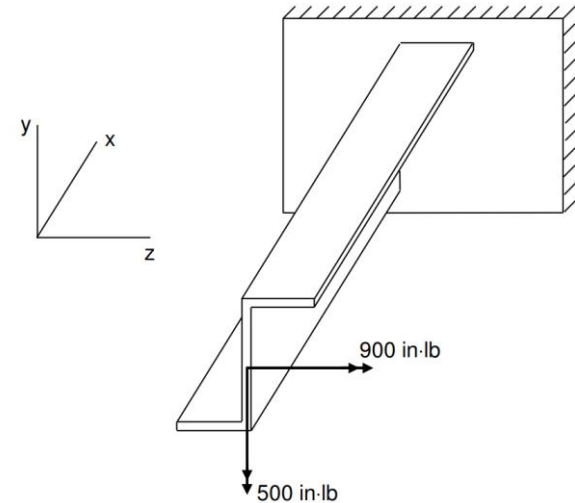
- Textbook outlines four steps to accomplish this:**
 - Step 1: Choose axis directions
 - Step 2: Locate centroid
 - Step 3: Place coordinate frame at the centroid and find inertia (MOI/POI) values
 - Step 4: Find values and signs for internal moment components

Textbook fails to mention this!



4.1: General Flexure Formula

- **STEP 1:** Textbook suggests choosing x-axis pointing into the page and y-axis pointing up → z-axis will point to the right
- **STEP 2:** To Locate the centroid, define temporary y and z axes as follows:



$$t = 0.064"$$



4.1: General Flexure Formula

- STEP 2: If we number each of the 3 segments making up the cross section as shown on the previous slide, this yields:**

$$\bar{y} = \frac{\bar{y}_1 A_1 + \bar{y}_2 A_2 + \bar{y}_3 A_3}{A_1 + A_2 + A_3}$$

$$\bar{y} = \frac{\left\{ \begin{aligned} &\overbrace{(1'' - 0.032'') \left[(0.064'') \left(\frac{9}{16}'' \right) \right]}^{0.036 \text{ in}^2} \\ &+ \left[\left(\frac{1'' - 2(0.064'')}{2} \right) + (0.064'') \right] \overbrace{\left[(1'' - 2(0.064'')) (0.064'') \right]}^{0.05581 \text{ in}^2} \\ &+ (0.032'') \overbrace{\left[(0.064'') \left(\frac{9}{16}'' \right) \right]}^{0.036 \text{ in}^2} \end{aligned} \right\}}{\underbrace{0.036 \text{ in}^2 + 0.05581 \text{ in}^2 + 0.036 \text{ in}^2}_{0.1278 \text{ in}^2}}$$

$\bar{y} = 0.5''$

$$\bar{z} = \frac{\bar{z}_1 A_1 + \bar{z}_2 A_2 + \bar{z}_3 A_3}{A_1 + A_2 + A_3}$$

$$\bar{z} = \frac{\left\{ \begin{aligned} &\left[\left(\frac{9}{16}'' - 0.064'' \right) + \frac{9}{32}'' \right] (0.036 \text{ in}^2) \\ &+ \left(\frac{9}{16}'' - 0.032'' \right) (0.05581 \text{ in}^2) + \left(\frac{9}{32}'' \right) (0.036 \text{ in}^2) \end{aligned} \right\}}{0.1278 \text{ in}^2}$$

$\bar{z} = 0.5305''$

4.1: General Flexure Formula

- STEP 3:** Here we make use of the parallel axis theorem and the fact that MOIs are additive:

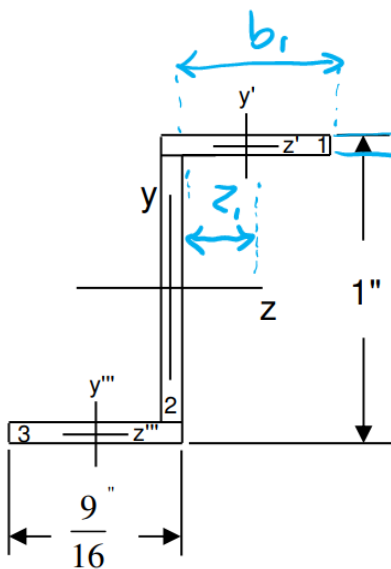
$$I_y = (I_y)_1 + (I_y)_2 + (I_y)_3$$

$$(I_y)_i = (I_{y'})_i + A z_i^2 = \frac{h_i b_i^3}{12} + A z_i^2$$

Handwritten calculations for the first term:

$$\frac{0.064''}{(b_1 \times h_1)} \rightarrow \left(\frac{9''}{16}\right)^3$$

$$\left(\frac{9}{16} \times \frac{1}{2}\right) - \frac{0.064}{2}$$



$t = 0.064''$

4.1: General Flexure Formula

- STEP 3: Continuing for I_{y2} and I_{y3} and summing the 3 MOIs yields:**

$$\begin{aligned}
 I_y = (I_y)_1 + (I_y)_2 + (I_y)_3 &= \left\{ \frac{\overbrace{0.0009492 \text{ in}^4}^{(0.064'')\left(\frac{9}{16}''\right)^3}}{12} + (0.036 \text{ in}^2) \left[\frac{\overbrace{0.2493''}{\frac{9}{32}'' - 0.032''}}{32} - 0.032'' \right]^2 \right\} \Rightarrow I_y = 0.006392 \text{ in}^4 \\
 &+ \left\{ \frac{[1'' - 2(0.064'')](0.064'')^3}{12} + 0 \right\} \\
 &+ \left\{ 0.0009492 \text{ in}^4 + (0.036 \text{ in}^2)(-0.2493'')^2 \right\}
 \end{aligned}$$



4.1: General Flexure Formula

- STEP 3: Proceeding in the same way for I_z yields:

$$I_z = (I_z)_1 + (I_z)_2 + (I_z)_3$$

Handwritten notes and diagram for $(I_z)_1$:

$(I_z)_1 = (I_{z_1}) + A y_1^2 = \frac{b_1 h_1^3}{12} + A y_1^2$

Diagram shows a rectangular area with width $b_1 = \frac{9}{16}$ inches and height $h_1 = 0.064$ inches. The centroidal axis is at $y_1 = \frac{1}{2} \times 1 - (\frac{1}{2} \times 0.064)$ inches.

$$I_z = (I_z)_1 + (I_z)_2 + (I_z)_3 = \left\{ \frac{0.00001229 \text{ in}^4}{12} + \left(\frac{9}{16} \text{ in} \right) (0.064 \text{ in})^3 + (0.036 \text{ in}^2) \left[\frac{0.468 \text{ in}}{0.5 \text{ in} - 0.032 \text{ in}} \right]^2 \right\} \Rightarrow I_z = 0.01933 \text{ in}^4$$

$$+ \left\{ \frac{(0.064 \text{ in}) [1 \text{ in} - 2(0.064 \text{ in})]^3}{12} + 0 \right\} + \left[0.00001229 \text{ in}^4 + (0.036 \text{ in}^2) (-0.468 \text{ in})^2 \right]$$



4.1: General Flexure Formula

- STEP 3: And for I_{yz} :

$$I_{yz} = (I_{yz})_1 + (I_{yz})_2 + (I_{yz})_3$$

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$$(I_{yz})_1 = (\cancel{I_{y'z'}})_1 + \underbrace{A}_\downarrow \underbrace{z}_\downarrow \underbrace{y}_\downarrow$$

Sym.

Already Have Been Calculated
for I_y and I_z in previous parts

$$I_{yz} = (I_{yz})_1 + (I_{yz})_2 + (I_{yz})_3 = \left[0 + (0.036 \text{ in}^2)(0.468'')(0.2493'') \right] + 0$$

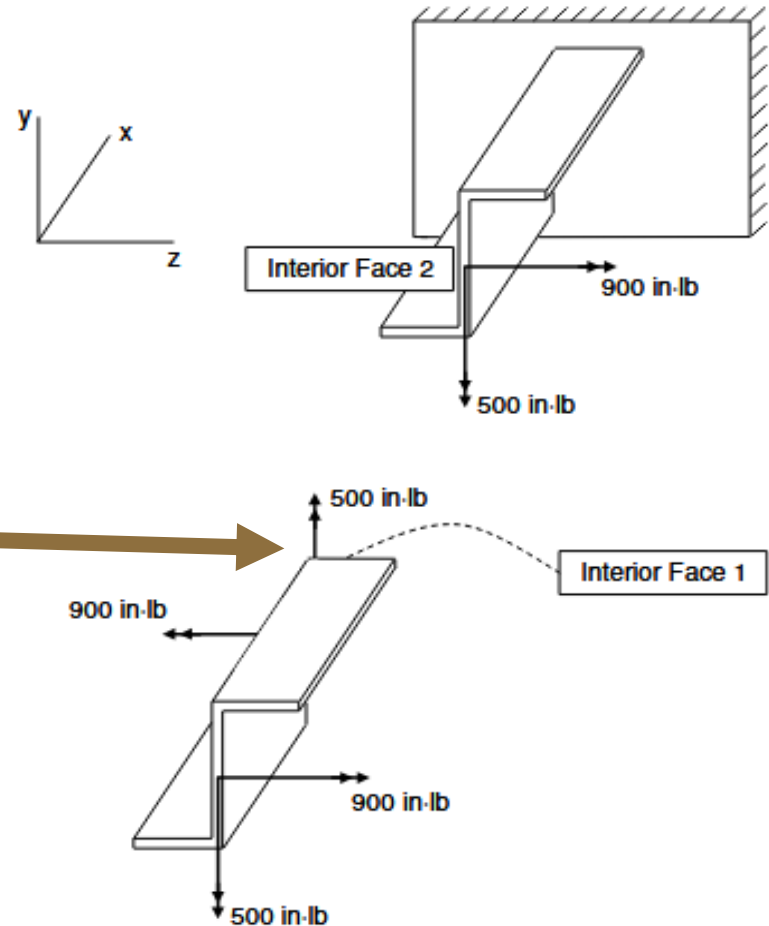
$$+ \left[0 + (0.036 \text{ in}^2)(-0.468'')(-0.2493'') \right]$$



$$I_{yz} = 0.008400 \text{ in}^4$$

4.1: General Flexure Formula

- **STEP 4:** In the problem statement (“Find the normal stress variation over the cross-section”), Dr Radosta failed to specify which cross-section, i.e. which internal/external face of the beam to analyze
- It turns out he meant this one (labeled “Interior Face 1”)
- Here $M_y = 500 \text{ in}\cdot\text{lb}$ and $M_z = -900 \text{ in}\cdot\text{lb}$



4.1: General Flexure Formula

- We now have all the values we need to insert into the General Flexure Formula:

$$\sigma_x = \frac{-y(I_y M_z + I_{yz} M_y) + z(I_{yz} M_z + I_z M_y)}{I_y I_z - (I_{yz})^2} = \frac{\left\{ -y \left[(0.006392 \text{ in}^4)(-900 \text{ in} \cdot \text{lb}) + (0.008400 \text{ in}^4)(+500 \text{ in} \cdot \text{lb}) \right] \right.}{(0.006392 \text{ in}^4)(0.01933 \text{ in}^4) - (0.008400 \text{ in}^4)^2} \left. + z \left[(0.008400 \text{ in}^4)(-900 \text{ in} \cdot \text{lb}) + (0.01933 \text{ in}^4)(+500 \text{ in} \cdot \text{lb}) \right] \right\}$$

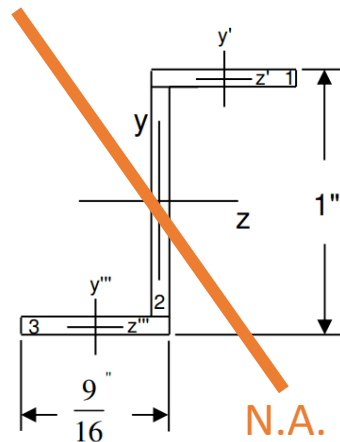
$\Rightarrow$

$$\sigma_x = (29,300)y + (39,720)z$$

- This answer makes sense because:**
 - The upper right portion of the cross section (when viewing the beam from the free end) is predominantly made up of “positive y, positive z” area
 - For “Interior Face 1” this means $\sigma_x > 0 \rightarrow$ tension
 - The lower left portion of the cross section is predominantly made up of “negative y, negative z” area $\rightarrow \sigma_x < 0 \rightarrow$ compression
 - If we had chosen “Interior Face 2” to calculate the stress distribution, these signs would be reversed

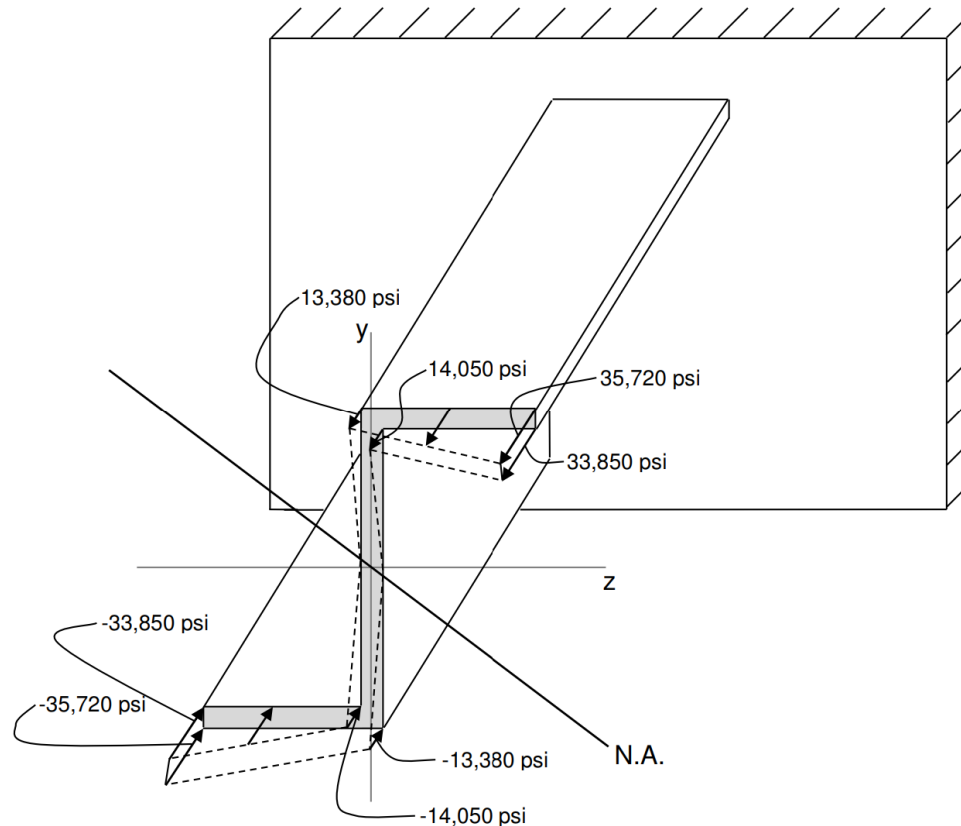
4.1: General Flexure Formula

- For part (b), the **neutral axis** is the axis in the cross-section of a beam along which there are no longitudinal stress or strain (as Alex would tell us 😊)
- Setting $\sigma_x = 0$ yields $0 = 29300y + 39720z \rightarrow y = -1.356z$
- N.A. therefore makes an angle of $\tan^{-1}(-1.356) = -53.39^\circ$ with the z axis; when viewed from the free end (i.e. along the positive x direction), it looks like this:



4.1: General Flexure Formula

- For part (c), the plot of $\sigma_x = 29300y + 39720z$ is a surface that looks like this:



- Question: where is the “ σ critical point” of the cross section, i.e. where is σ_x a maximum? And what is the maximum value?

4.1: General Flexure Formula

- We see that σ_x varies linearly in both y and z
- In this case the critical point is easy to find because the location of max y is also the location of max z : $(0.5 \text{ in}, 0.5305 \text{ in})$, i.e. the top right corner of the cross-section
- Max σ_x here is 35721.46 psi
- The bottom left corner of the cross-section $(-0.5 \text{ in}, -0.5305 \text{ in})$ is also a critical point ($\sigma_x = -35721.46 \text{ psi}$)

